**Assignment 3. Gauss-Jordan, LU & Cholesky** Marks 20

Posted on 20.08.2026 @ 2:30 pm and due on 20.08.2026 @ 5:40 pm

---

1. Prepare library routines for forward-backward solution and Cholesky decomposition, which must include a check for symmetric matrix. **[5+5]**

2. Given below is a system of linear equations. Use the LU decomposition to solve it by forward-backward substitution. Use either *Doolittle* or *Crout*. **[5+5]**

$$\begin{aligned} 19 &= a_1 - a_2 + 4a_3 + 2a_5 + 9a_6 \\ 2 &= 5a_2 - 2a_3 + 7a_4 + 8a_5 + 4a_6 \\ 13 &= a_1 + 5a_3 + 7a_4 + 3a_5 - 2a_6 \\ -7 &= 6a_1 - a_2 + 2a_3 + 3a_4 + 8a_6 \\ -9 &= -4a_1 + 2a_2 + 5a_4 - 5a_5 + 3a_6 \\ 2 &= 7a_2 - a_3 + 5a_4 + 4a_5 - 2a_6 \end{aligned}$$

3. Use Cholesky factorization for the matrix $\mathbb{A}$ and solve for $\mathbb{A}\mathbf{x} = \mathbf{b}$ for the following **[5]**

$$\mathbb{A} = \begin{pmatrix} 4 & 1 & 1 & 1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 2 & 0 \\ 1 & 1 & 0 & 2 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 3 \\ 3 \\ 1 \\ 3 \end{pmatrix}$$